

Banking and Monetary Policy

Chapter 20

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

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IN THIS CHAPTER, YOU WILL LEARN:

- ▶ How banks “create” money
- ▶ How the Federal Reserve Bank controls the money supply

Money supply measures, January 2021

Symbol	Assets included	Amount in Oct. 2023 (billions of dollars)
<i>C</i>	Currency	\$2,324.9
<i>M1</i>	Currency plus demand deposits, traveler's checks, and other checkable deposits	18,034.3
<i>M2</i>	<i>M1</i> plus retail money market mutual fund balances, saving deposits (including money market deposit accounts), and small time deposits	20,671.1

Banks' role in the monetary system, part 1

Recall from Chapter 5 that we define the money supply (M) as equal to currency (C) plus demand (checking account) deposits (D):

- ▶ $M = C + D$
- ▶ The central bank can control the money supply, but since the money supply includes demand deposits, the banking system also plays an important role.



FOMC meeting in the Eccles Building boardroom, 2016 (Yellen chairing). This is where monetary policy is set.

Wikimedia, PD-US Gov.

A few preliminaries

- ▶ **Reserves** (R): the portion of deposits that banks have not lent.
- ▶ A bank's liabilities include deposits; assets include reserves and outstanding loans.
- ▶ **100-percent-reserve banking**: a system in which banks hold all deposits as reserves.
- ▶ **Fractional-reserve banking**: a system in which banks hold a fraction of their deposits as reserves.

Banks' role in the monetary system, part 2

To understand the role of banks, we will consider three scenarios:

- ▶ No banks
 - ▶ 100-percent-reserve banking (banks hold all deposits as reserves)
 - ▶ Fractional-reserve banking (banks hold a fraction of deposits as reserves and use the rest to make loans)
- ▶ In each scenario, we assume currency, $C = \$1,000$.

Scenario 1: No banks

- ▶ With no banks, $D = 0$ and $M = C = \$1,000$.
- ▶ This is easy: The money supply is equal to the amount of currency in circulation.

Scenario 3: Fractional-reserve banking (1 of 4)

- ▶ Suppose banks hold 20% of deposits in reserve ($rr = \mathbf{0.2}$), making loans with the rest.
- ▶ Firstbank will make \$800 in loans.
- ▶ **FIRSTBANK'S balance sheet**
- ▶ The money supply now equals \$1,800:
 - ▶ Depositor has \$1,000 in demand deposits.
 - ▶ Borrower holds \$800 in currency.

Scenario 3: Fractional-reserve banking (2 of 4)

- ▶ Suppose banks hold 20% of deposits in reserve, making loans with the rest.
- ▶ Firstbank will make \$800 in loans.
- ▶ **FIRSTBANK'S balance sheet**
- ▶ LESSON: In a fractional-reserve banking system, banks create money.
- ▶ The money supply now equals \$1,800:
 - ▶ Depositor has \$1,000 in demand deposits.
 - ▶ Borrower holds \$800 in currency.

Scenario 3: Fractional-reserve banking (3 of 4)

- ▶ Suppose the borrower deposits the \$800 in Secondbank.
- ▶ Initially, Secondbank's balance sheet is:
- ▶ **SECONDBANK'S balance sheet**
- ▶ Secondbank will loan 80% of this deposit and keep 20% in reserve.

Scenario 3: Fractional-reserve banking (4 of 4)

- ▶ If this \$640 is eventually deposited in Thirdbank,
- ▶ Then Thirdbank will keep 20% of it in reserve and loan out the rest:
- ▶ **THIRDBANK'S balance sheet**

Finding the total amount of money

- ▶ Original deposit = \$1,000
- ▶ + Firstbank lending = \$ 800
- ▶ + Secondbank lending = \$ 640
- ▶ + Thirdbank lending = \$ 512
- ▶ + other lending . . .
- ▶ Adding up all the lending yields the following formula:
- ▶ Total money supply = $(1/rr) \times \$1,000$ where rr = ratio of reserves to deposits
- ▶ In our example, $rr = 0.2$, so $M = \$5,000$

Money creation in the banking system

- ▶ *A fractional-reserve banking system creates money, but it doesn't create wealth.*
- ▶ *Bank loans give borrowers some new money*
- ▶ *and an equal amount of new debt.*

Bank capital, leverage, and capital requirements, part 1

- ▶ **Bank capital:** the resources a bank's owners have put into the bank; difference between the value of a bank's assets and liabilities
- ▶ A more realistic balance sheet:

Bank capital, leverage, and capital requirements, part 2

- ▶ **Leverage**: the use of borrowed money to supplement existing funds for purposes of investment
- ▶ *Leverage ratio* = assets/capital
- ▶ = $$(200 + 500 + 300)/\$50 = \mathbf{20}$

Bank capital, leverage, and capital requirements, part 3

- ▶ Being highly leveraged makes banks vulnerable.
- ▶ Example: Suppose a recession causes our bank's loans and securities to fall by 5% to \$950.
- ▶ Then, $\text{capital} = \text{assets} - \text{liabilities} = 950 - 950 = 0$

Bank capital, leverage, and capital requirements, part 4

- ▶ What happens if the value of assets declines by more than 10 percent to 900?
- ▶ In this case, assets < liabilities
- ▶ Capital = assets - liabilities = 900 - 950 = -50
- ▶ Bank is insolvent.

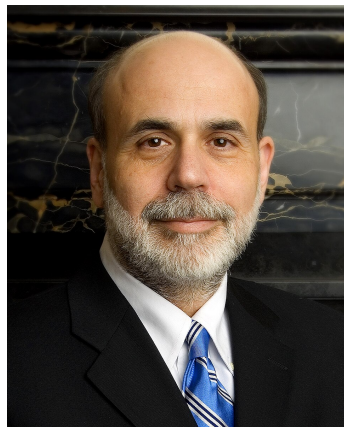
Bank capital, leverage, and capital requirements, part 5

Capital requirement:

- ▶ minimum amount of capital mandated by regulators
 - ▶ intended to ensure that banks will be able to pay off depositors
 - ▶ higher for banks that hold more risky assets
- ▶ 2008–2009 financial crisis:
 - ▶ Losses on mortgages shrank bank capital, slowed lending, and exacerbated the recession.
 - ▶ Regulatory changes imposed higher capital requirements with the goal of reducing the likelihood of future crisis.

Bank runs: Liquidity versus solvency, part 1

- ▶ A bank run occurs when a large number of depositors withdraw their deposits over concerns of the bank's solvency.
- ▶ A bank's deposits are liquid, whereas a bank's assets (loans) are illiquid.
- ▶ When an unanticipated large fraction of depositors demand their deposits, a bank can face a liquidity crisis as it cannot easily turn its assets into cash or liquidate them.
- ▶ The bank may need a bailout if liquidity is the only problem.



Ben S. Bernanke,
Fed Chair 2006–2014.
Great-Depression scholar
who ran the 2008 rescue
that averted a repeat.

Bank runs: Liquidity versus solvency, part 2

- ▶ Solvency is a more serious problem.
- ▶ When a bank is insolvent, the value of its assets falls well below the value of its liabilities.
- ▶ Not enough resources to repay all depositors even if all of the assets are sold.
- ▶ Solution: Bank closure or bank regulators take the bank over and provide the necessary liquidity to make it solvent.

CASE STUDY: The bank failures of 2023, part 1

- ▶ Silicon Valley Bank (SVB) experienced rapid deposit growth from its clients, mostly startups and tech ventures.
- ▶ By 2022, it was the 18th largest bank in the United States.
- ▶ A large proportion of its depositors' money was invested in long-term government bonds.
- ▶ As inflation surged in 2022, the value of these long-term bonds fell significantly, severely impacting SVB's assets.

CASE STUDY: The bank failures of 2023, part 2

- ▶ Many depositors were not insured by the FDIC, because they exceeded the 250k insurance limit.
- ▶ Rumors started to spread over social media that SVB was suffering large losses on its assets.
- ▶ Rapid withdrawals in early 2023 led to SVB's inability to meet depositors' demand.
- ▶ Regulators took over the bank in March 2023, and the FDIC and U.S. Treasury decided to insure all deposits to prevent a banking panic.

CASE STUDY: The bank failures of 2023, part 3

- ▶ Following SVB's collapse, other banks such as Signature Bank and First Republic Bank also faced a similar crisis.
- ▶ Questions were raised about the risk that these banks undertook with depositors' money and the oversight failures of regulators

A model of the money supply

- ▶ Exogenous variables
- ▶ **Monetary base:** *controlled by the central bank*
- ▶ **Reserve-deposit ratio:** *depends on regulations and bank policies*
- ▶ **Currency-deposit ratio:** *depends on households' preferences*

Solving for the money supply (1 of 4)

Some notations:

- ▶ M = money supply
- ▶ B = monetary base
- ▶ C = currency
- ▶ D = demand deposits
- ▶ R = bank reserves
- ▶ $cr = C/D$ = currency–deposit ratio
- ▶ $rr = R/D$ = reserve–deposit ratio

By definition:

$$M = C + D, \quad B = C + R.$$

So the money-supply / monetary-base ratio is

$$\frac{M}{B} = \frac{C + D}{C + R}.$$

Solving for the money supply (2 of 4)

Divide the numerator and denominator of M/B by D :

$$\frac{M}{B} = \frac{C+D}{C+R} = \frac{\frac{C}{D} + 1}{\frac{C}{D} + \frac{R}{D}} = \frac{cr + 1}{cr + rr}.$$

Solving for the money supply (3 of 4)

Denoting

$$m \equiv \frac{cr + 1}{cr + rr},$$

we get

$$\boxed{M = m \times B}.$$

The money supply equals the monetary base multiplied by m , the **money multiplier**.

Solving for the money supply (4 of 4)

- ▶ If $rr < 1$, then $m > 1$: each \$1 of monetary base creates more than \$1 of money.
- ▶ If the monetary base changes by ΔB , then

$$\Delta M = m \times \Delta B.$$

- ▶ m is the **money multiplier** — the increase in the money supply from a one-dollar increase in the monetary base.

▶ Numerical example

NOW YOU TRY: The money multiplier

1. Suppose households decide to hold more of their money as currency and less in the form of demand deposits.
 - ▶ Determine the impact on the money supply.
 - ▶ Explain the intuition for your result.

NOW YOU TRY: The money multiplier, solution

1. Impact of an increase in the currency–deposit ratio $\Delta cr > 0$.
 - ▶ An increase in cr increases the denominator of m proportionally more than the numerator. So m falls, causing M to fall.
 - ▶ If households deposit less of their money, then banks can't make as many loans, so the banking system won't be able to create as much money.

The instruments of monetary policy, part 1

The Fed can change the monetary base by using:

- ▶ open market operations (the Fed's preferred method of monetary control)
 - ▶ To increase the base, the Fed could buy government bonds, paying with new dollars.
 - ▶ the **discount rate**: the interest rate the Fed charges on loans to banks
 - ▶ To increase the base, the Fed could lower the discount rate, encouraging banks to borrow more reserves.

The instruments of monetary policy, part 2

The Fed can change the reserve–deposit ratio by using:

- ▶ **reserve requirements:** Fed regulations impose a minimum reserve–deposit ratio.
 - ▶ To reduce the reserve–deposit ratio, the Fed could reduce reserve requirements.
 - ▶ **interest on reserves:** The Fed pays interest on bank reserves deposited with the Fed.
 - ▶ To reduce the reserve–deposit ratio, the Fed could pay a lower interest rate on reserves.

Why the Fed can't precisely control M

- ▶ Households can change cr , causing m and M to change.
- ▶ Banks often hold **excess reserves** (reserves above the reserve requirement).
- ▶ If banks change their excess reserves, then rr , m , and M change.

CASE STUDY: Bank failures in the 1930s, part 1

From 1929 to 1933:

- ▶ more than 9,000 banks closed
 - ▶ the money supply fell 28 percent
- ▶ This drop in the money supply may not have caused the Great Depression, but it certainly contributed to its severity.

CASE STUDY: Bank failures in the 1930s, part 2

- ▶ Loss of confidence in banks: increases cr , reduces m
- ▶ Banks became more cautious: increases rr , reduces m

CASE STUDY: Bank failures in the 1930s, part 3

	August 1929	March 1933	% change
Money Supply	26.5	19.0	-28.3%
Currency	3.9	5.5	41.0
Demand Deposit	22.6	13.5	-40.3
Monetary Base	7.1	8.4	18.3
Currency	3.9	5.5	41.0
Reserves	3.2	2.9	-9.4
Money Multiplier	3.7	2.3	-37.8
Reserve-deposit ratio	0.14	0.21	50.0
Currency-deposit ratio	0.17	0.41	141.2

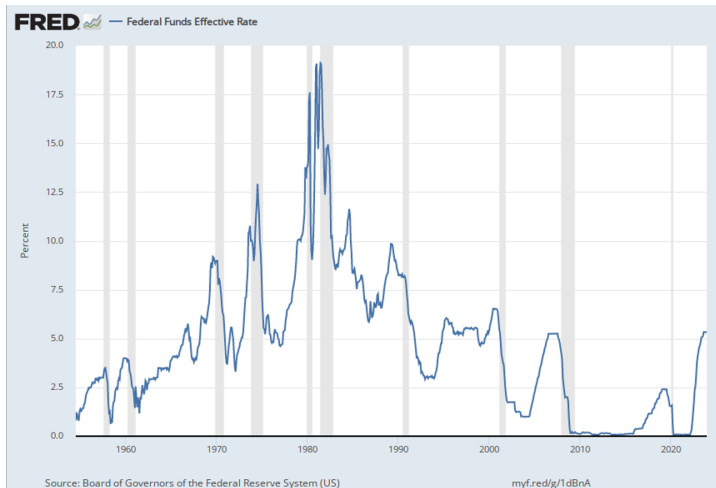
Could this happen again?

- ▶ Many policies have been implemented since the 1930s to prevent such widespread bank failures.
- ▶ An example is federal deposit insurance to prevent bank runs and large swings in the currency–deposit ratio.

The Fed's current monetary regime, part 1

- ▶ Monetary policy is set by the FOMC, which meets every six weeks to set a target for the Federal Funds Rate (FFR).
- ▶ FFR is the overnight rate at which banks lend reserves to each other.
- ▶ FFR influences other short-term rates and ultimately economic conditions.

Federal funds rate, 1954–2023



The Fed's current monetary regime, part 2

- ▶ FFR is determined by the demand and supply for overnight loans from banks.
- ▶ Historically, the Fed used open-market operations (OMO) to get the actual federal funds rate as close as possible to its target.
- ▶ Since the financial crisis of 2008–2009, the FED began paying interest on banks' reserves.
- ▶ Since then, it has relied on the interest rate it pays on reserves (not OMO) to influence the FFR.

The Fed's current monetary regime, part 3

As discussed in Chapter 5, in recent years, other than setting a target for the FFR, the Fed has also used unconventional monetary policy when the FFR was close to its zero-lower bound:

- ▶ forward guidance (commits to a future path for the FFR)
 - ▶ quantitative easing (purchasing a broader range of securities to increase liquidity)

CHAPTER SUMMARY (1 of 3)

- ▶ Fractional reserve banking creates money because each dollar of reserves generates many dollars of demand deposits.
- ▶ The money supply depends on the:
 - ▶ monetary base
 - ▶ currency–deposit ratio
 - ▶ reserve ratio

CHAPTER SUMMARY (2 of 3)

- ▶ Bank capital, leverage, and capital requirements
 - ▶ Bank capital is the owners' equity in the bank.
 - ▶ Because banks are highly leveraged, a small decline in the value of bank assets can have a huge impact on bank capital.
 - ▶ Bank regulators require that banks hold sufficient capital to ensure that depositors can be repaid.

CHAPTER SUMMARY (3 of 3)

The Fed can control the money supply with:

- ▶ open-market operations
 - ▶ the reserve requirement
 - ▶ the discount rate
 - ▶ interest on reserves

Appendix: Money multiplier — a numerical example

Recall: $m = \frac{cr + 1}{cr + rr}$, $M = m \cdot B$.

Baseline (typical U.S. 1980s). Currency-deposit ratio $cr = 0.4$, reserve-deposit ratio $rr = 0.1$, monetary base $B = 800$ (billion \$).

$$m = \frac{0.4 + 1}{0.4 + 0.1} = \frac{1.4}{0.5} = 2.8,$$

$$M = 2.8 \times 800 = 2,240.$$

So each \$1 the Fed creates translates to \$2.80 of money supply through the banking system.

[◀ back](#)

Appendix — how a financial crisis shrinks the multiplier

Panic scenario. Depositors pull cash out: cr jumps from 0.4 to 0.8. Nervous banks hold more precautionary reserves: rr rises from 0.1 to 0.2. Fed holds B constant.

$$m_{\text{new}} = \frac{0.8 + 1}{0.8 + 0.2} = \frac{1.8}{1.0} = 1.8,$$

$$M_{\text{new}} = 1.8 \times 800 = 1,440.$$

Money supply falls by $2,240 - 1,440 = 800$ — a 36% contraction with no policy change.

This is (part of) what happened in the Great Depression: M_2 fell about 30% between 1929 and 1933 mostly because cr and rr rose sharply — *not* because the Fed cut B . Bernanke's Fed in 2008 responded by expanding B massively to offset the multiplier collapse.

◀ back

Appendix — QE and the collapse of m after 2008

The Fed pushed B from \$0.9T (2007) to \$4.5T (2014). Did M_2 rise proportionally?

Year	Monetary base (B)	M2	Implied $m = M_2/B$
2007	0.85T	7.5T	8.8
2014	4.0T	11.6T	2.9
2022	5.7T	21.5T	3.8

The multiplier *collapsed* from about 9 to 3 — banks parked reserves at the Fed instead of lending them out (rr effectively rose sharply). This is why massive QE didn't produce runaway inflation the way monetarists (e.g., Meltzer) feared.

[◀ back](#)